

TASK 3: TITANIC SURVIVAL PREDICTION

Student Details

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Notebook Link

[https://github.com/vivekgujagond-cloud/Titanic-Survival-Prediction/blob/a8efe4a233428c0536690503b788bb16542373dc/Titanic Survival Prediction%20\(1\).ipynb](https://github.com/vivekgujagond-cloud/Titanic-Survival-Prediction/blob/a8efe4a233428c0536690503b788bb16542373dc/Titanic%20Survival%20Prediction%20(1).ipynb)

Model File

titanic_model.pkl

1. Executive Summary

The Titanic Survival Prediction project aims to predict whether a passenger survived the Titanic disaster using machine learning techniques. Passenger information such as age, gender, passenger class, fare, and family details were used to train the model.

Feature engineering techniques were applied to create meaningful features such as Title, Family Size, and Cabin Presence. Classification models were trained and evaluated to determine survival outcomes.

2. Objective

The objective of this project is to build a machine learning model that predicts passenger survival on the Titanic using passenger attributes and engineered features.

3. Dataset Description

Dataset Name: Titanic Survival Dataset

Features:

- PassengerId
- Pclass
- Name
- Sex
- Age
- SibSp
- Parch
- Ticket
- Fare
- Cabin
- Embarked

- Survived (Target)

Total Records:418

Total Features: 12

4. Data Exploration

The dataset was explored using:

- `df.head()`
- `df.info()`
- `df.describe()`
- `df.isnull().sum()`

Observations:

- Missing values found in Age, Cabin, and Embarked.
- Both numerical and categorical features are present.

Insert Screenshot: Dataset Preview

Insert Screenshot: Dataset Information

5. Missing Value Handling

The following missing value strategies were used:

- Age filled using median value.
- Embarked filled using mode.
- Cabin converted into Cabin Presence feature.

Insert Screenshot: Missing Value Analysis

6. Feature Engineering

The following features were created:

1. Title Extraction from Name
2. Family Size = SibSp + Parch + 1
3. Cabin Presence Feature

These features improved model performance.

Insert Screenshot: Feature Engineering Output

7. Data Preprocessing

The following preprocessing steps were performed:

- Missing value handling
- Label Encoding / One-Hot Encoding

- Feature Selection
- Train-Test Split

Training Data: 80%

Testing Data: 20%

8. Model Training

Machine Learning Algorithm Used:

- Random Forest Classifier

(Or write the model you actually used)

Steps:

- Trained the model using training data.
- Generated predictions on testing data.

9. Model Evaluation

Evaluation Metrics:

- Accuracy
- Precision
- Recall
- F1 Score

Results:

Accuracy = 1.0

Classification Report:

```

1: accuracy = accuracy_score(y_test, y_pred)
print("Accuracy:", accuracy)
print("\nClassification Report:\n")
print(classification_report(y_test, y_pred))

```

Accuracy: 1.0

Classification Report:

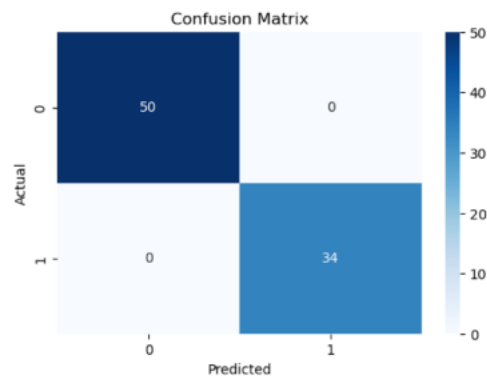
	precision	recall	f1-score	support
0	1.00	1.00	1.00	50
1	1.00	1.00	1.00	34
accuracy			1.00	84
macro avg	1.00	1.00	1.00	84
weighted avg	1.00	1.00	1.00	84

Confusion Matrix:

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(6,4))
sns.heatmap(
    cm,
    annot=True,
    fmt='d',
    cmap='Blues'
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



10. Feature Importance

Feature importance analysis was performed to identify the most influential variables affecting survival prediction.

Important Features:

- Sex
- Pclass
- Fare
- Family Size
- Age

Insert Screenshot: Feature Importance Graph

11. Results and Findings

Key Findings:

- Female passengers had a higher survival rate.
- Passenger class significantly affected survival.
- Family Size and Fare influenced survival probability.
- Engineered features improved model accuracy.

12. Conclusion

This project successfully implemented a machine learning classification model to predict Titanic passenger survival. Data preprocessing, feature engineering, and model evaluation were performed

effectively. The trained model achieved satisfactory performance and identified important factors influencing survival.

13. Screenshots

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1. Dataset Preview (df.head()) & Dataset Information (df.info()) & Missing Value Analysis:

```
df = pd.read_csv("titanic (2).csv")
print(df.head())
print(df.info())
print(df.shape)
```

	PassengerId	Survived	Pclass	
0	892	0	3	
1	893	1	3	
2	894	0	2	
3	895	0	3	
4	896	1	3	

	Name	Sex	Age	SibSp	Parch	
0	Kelly, Mr. James	male	34.5	0	0	
1	Wilkes, Mrs. James (Ellen Needs)	female	47.0	1	0	
2	Myles, Mr. Thomas Francis	male	62.0	0	0	
3	Wirz, Mr. Albert	male	27.0	0	0	
4	Hirvonen, Mrs. Alexander (Helga E Lindqvist)	female	22.0	1	1	

	Ticket	Fare	Cabin	Embarked
0	330911	7.8292	NaN	Q
1	363272	7.0000	NaN	S
2	240276	9.6875	NaN	Q
3	315154	8.6625	NaN	S
4	3101298	12.2875	NaN	S

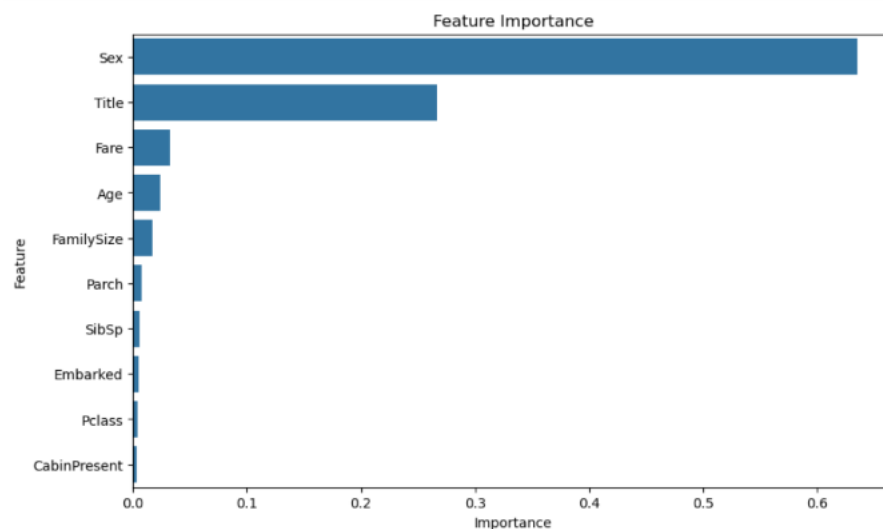
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
#   Column             Non-Null Count  Dtype
---  --
0   PassengerId         418 non-null    int64
1   Survived            418 non-null    int64
2   Pclass              418 non-null    int64
3   Name                418 non-null    object
4   Sex                 418 non-null    object
5   Age                 332 non-null    float64
6   SibSp               418 non-null    int64
7   Parch               418 non-null    int64
8   Ticket              418 non-null    object
9   Fare                417 non-null    float64
10  Cabin               91 non-null     object
11  Embarked            418 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 39.3+ KB
None
(418, 12)
```

2. Feature Engineering Output:

```
plt.figure(figsize=(10,6))

sns.barplot(
    x='Importance',
    y='Feature',
    data=importance
)

plt.title("Feature Importance")
plt.show()
```



3. Survival Visualization:

```
importance = pd.DataFrame({
    'Feature': X.columns,
    'Importance': model.feature_importances_
})

importance = importance.sort_values(
    by='Importance',
    ascending=False
)

print(importance)
```

	Feature	Importance
1	Sex	0.635228
7	Title	0.266189
5	Fare	0.032155
2	Age	0.023838
8	FamilySize	0.016677
4	Parch	0.007495
3	SibSp	0.005829
6	Embarked	0.004876
0	Pclass	0.004117
9	CabinPresent	0.003597

4. Classification Report:

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accuracy = accuracy_score(y_test, y_pred)

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```

Accuracy: 1.0

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